

SHIVARAMAN T

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Profile Summary

B.Tech Electronics and Communication Engineering student specializing in AI & Cybernetics with strong fundamentals in **engineering analysis, electronics, control systems, sensors, measurement, and system validation**. Hands-on experience in **industrial hardware testing, data acquisition, troubleshooting, simulation tools, system integration, and technical documentation**, with strong interest in **CAE, engineering simulation, product development, and industrial applications**.

Education

Vellore Institute of Technology

Sept 2023 – May 2027

B.Tech in Electronics and Communication Engineering (AI & Cybernetics)

CGPA: **8.68 / 10**

Technical Skills

Engineering Analysis & Simulation:	Engineering Analysis, Engineering Fundamentals, Simulation, Measurement, Testing, Validation, Data Acquisition, Result Interpretation, MATLAB, Simulink
CAE-Relevant Fundamentals:	Finite Element Analysis Fundamentals, Engineering Mechanics, Stress and Strain Fundamentals, Material Properties, Loads and Boundary Conditions, Structural Analysis Fundamentals, Thermal Analysis Fundamentals
Electronics & Control:	Circuit Analysis, Analog Electronics, Digital Electronics, Power Electronics, Control Systems, Feedback Systems, PID Control, Sensors, Signal Conditioning, Calibration
Hardware & Embedded:	ESP32, Arduino, Microcontrollers, Embedded C, UART, I2C, SPI, PWM, ADC, GPIO, Motor Control, Relay Control, Industrial Automation, System Integration
Engineering & Design Tools:	Cadence Allegro, OrCAD Capture CIS, Keysight ADS, LTspice, MATLAB, Simulink, Proteus, PCB Design, PCB Layout, Schematic Analysis, Design Validation

Experience

Zettaone Technologies, Bengaluru — PCB Design Intern

Jun 2026 – Jul 2026

- Worked on **hardware engineering and engineering design** using Cadence Allegro and OrCAD Capture CIS, including schematic analysis, component placement, PCB layout, stackup, decoupling, trace-width selection, and design-rule checks.
- Performed **simulation, measurement, testing, and troubleshooting** using ADS and LTspice; analyzed impedance, switching waveforms, DC/AC behavior, and validated circuit performance against design requirements.

Aqualis Inspection Services, Dubai — Engineering Intern

May 2025 – Jun 2025

- Assisted in inspection and validation of **5+ industrial hardware monitoring units**, performing sensor data acquisition, measurement, hardware testing, system-level troubleshooting, and engineering validation.
- Prepared **wiring diagrams, inspection documentation, test reports, and validation records** while supporting hardware fault analysis and system verification.

Projects

Industrial Machine Health Monitoring System — Machine Monitoring / Sensors / IoT

Present

- Designed a machine monitoring concept for measuring **vibration, temperature, current, and gas parameters** to support machine-condition assessment and industrial maintenance.
- Developed the system architecture for **sensor data acquisition, signal processing, condition monitoring, IoT communication, and result visualization** for engineering analysis.

Smart Aeroponics System — ESP32 / IoT / Closed-Loop Control

Apr 2026

- Developed an ESP32-based **monitoring and control system** for real-time measurement of pH, TDS, temperature, and water level using sensor interfacing, ADC, and 1-Wire communication.
- Implemented **closed-loop control, actuator integration, calibration, testing, and troubleshooting** using sensor feedback, relay drivers, peristaltic pumps, and an ultrasonic nebulizer.

Automatic Aerator System (AquaDox) — Arduino / Sensors / Control Systems

Sep 2024

- Developed an automated **dissolved oxygen monitoring and aeration control system** using microcontroller-based sensor acquisition and threshold-based control.
- Integrated **sensor calibration, signal conditioning, ADC interfacing, actuator control, firmware logic, testing, and hardware debugging** for reliable system operation.

Publication & Achievements

Smart Helmet Systems: IoT Enhanced Safety and Integrated Display for Two Wheelers — 6th International Conference on Recent Advancements in Mechanical Engineering (ICRAME 2025), NIT Silchar. Publication

- Global Rank 1347** — TCS CodeVita Season 13.
- Top 100** — Zealestra X AWS Machine Learning Challenge.
- Selected Participant** — Bharatiya Antariksh Hackathon 2024 (ISRO).

Certifications

Arduino Programming & Embedded Systems — GUVI-HCL | **MATLAB Signal Processing** | **Embedded Systems** — Raspberry Pi & Arduino | **AWS Certified Cloud Practitioner**